

UME743 : Computational Fluid Dynamics

L	T	P	Cr
2	0	2	3.0

Course Objective: This course introduces the basic knowledge of governing equations for fluid flow and different turbulence models. The course also introduces the concept of numerical methods used to solve the partial differential equation. Further, solve the fluid flow problem using CFD tool.

Syllabus

Introduction: Motivation and role of computational fluid dynamics, concept of modeling and simulation. Benefits and limitations of CFD software tools.

Governing equations of fluid dynamics: Continuity equation, momentum equation, energy equation, various simplifications, dimensionless equations and parameters, convective and conservation forms, incompressible flows, source panel method and vortex panel method. **Nature of equations:** Classification of PDE, general Types of parabolic, elliptic and hyperbolic equations, boundary and initial conditions.

Finite difference method: Discretization, various methods of finite differencing, stability, method of solutions.

Finite Volume methods: Integral Approach, discretization & Higher order scheme.

Turbulence modelling: Turbulence, effect of turbulence on N-S equations, different turbulent modelling scheme, Error and uncertainty.

Incompressible Viscous Flows: Stream function-vorticity formulation, solution for pressure, applications to internal flows and boundary layer flows

Assignment: Research assignment given to the students in group related to flow simulation of different NACA profile aerofoil section, diesel injector, heat exchanger etc. using available CFD tools.

Course Learning Objectives (CLO)

The students will be able to:

1. derive and analyse the various types of fluid flow governing equations.
2. analyse the internal fluid flow phenomena of thermal and fluid system.
3. Simulate engineering problems using commercial CFD tools.

Text Books

1. Ghosdastidar, P. S., Computer Simulation of Flow and Heat Transfer, McGraw Hill (1998)
2. Roache, P. J., Computational Fluid Dynamics, Hermosa (1998).
3. Wendt, J. F., Computational Fluid Dynamics An Introduction, Springer-Verlag (2008).

Reference Books

1. Muralidhar, K. and Sundararajan, T., Computational Fluid Flow and Heat Transfer, Narosa (2008) 2nd ed.
2. Jaluria, Y. and Torrance, K. E., Computational Heat Transfer, Taylor & Francis (2003).
3. Patankar, S. V., Numerical Heat Transfer and Fluid Flow, Taylor & Francis (2007).

Evaluation Scheme

S.No.	Evaluation Elements	Weightage (%)
1.	MST	25
2.	EST	40
3.	Sessional (May include Project/Quizes/Assignments Evaluation)	35